

Midterm Exam 2023 (Operating System)

題目不回收。2, 3為是非, 4複選。11題請附計算過程, 滿分:108

1. (4%)組別(Group ID)為 ____ (G?? or H??)
2. **F** (3%, T or F) In most cases, preemptive-priority scheduling produces less average waiting time than non-preemptive-priority scheduling.
大多數情形, preemptive-priority scheduling產生較少的average waiting time
SRT的平均等待時間小於SJF, 但是要推廣到大多數的priority scheduling會有困難, 讓等待時間變小的是”短的優先”, 而不是”搶佔”。如果長的工作較優先呢?
3. Process P performs the following operations one by one
 - (i) forks Process A,
 - (ii) creates an ordinary pipe H,
 - (iii) forks Process B
 - (iv) creates Thread C
 - (v) creates Thread D,Indicate true or false on the following statements, and briefly state the reasons.
T (3%, T or F) (a) The starting address of Process A and Process B are the same
F (3%, T or F) (b) The starting address of Thread C and Thread D are the same
F (3%, T or F) (c) Process A is able to communicate with Process B via Pipe H
T (3%, T or F) (d) Thread C is able to communicate with Thread D via Pipe H
pipe屬於process context, fork A之時, 並沒有H, 所以A看不到H, threads共享process context, 所以C, D可以透過H通信
fork 行程時, 連stack 內容也一併複製, 父子行程具有相同的歷史, 起點也相同, 就是main(). 但是create thread時, 並不複製stack, thread的起點就是給定function
4. **B** (6%, 複選, multiple choice) The NIC generates a signal to interrupt CPU _____. (a)whenever it receives a frame, (b)whenever it receives a frame destined for the host, (c)whenever is receives a CRC-error frame
CRC-error的訊框丟掉就好
5. (8%)There are two variants of priority scheduling, either preemptive or non-preemptive, can it be applied to RR or FCFS? priority scheduling有兩種版本, RR 與 FCFS 是否也有這兩種? 請敘述理由. **只有preemptive RR, 也只有non-preemptive FCFS, RR+priority會有兩種變形, 但是都屬於preemptive**
6. (8%)Compare open-source with closed-source software subject to (a)community support **open-source 較強** and (b)compatibility **closed-source 較強** 針對兩點來做比較
7. (8%)Modern Operating Systems are majorly implemented by “high level languages” or “assembly languages”? Please state two major considerations.現代作業系統的實作大多仰賴高階還是組合語言? 舉出兩點原因**高階, 易寫, 易debug, 易於了解, compiler的最佳化選項產生優質程式碼.. 重點不在對錯, 而是應該有的知識. 高階語言易寫, 易讀, 初學者也知道此事, OS非常重視效能, 為什麼沒有選擇組語?**
8. (8%) (a)What is the differences between “task parallelism” and “data parallelism”? (b)Show it by example, (c)Which one is more difficult to implement, state your reason. 兩種平行化的差異點, 舉例說明, 哪一種的實作較有困難**task parallelism實作較有困難**
9. (8%)Please give a practical instance show that the multi-threaded approach is more efficient than single-threaded approach, even there is only one CPU, one core. 可否舉出一個實例說明, 既使只有一個單核CPU, 採取multi-thread 的方式會比較(相對於single thread)有效率(=執行時間較短)。如果認為不可能,

也請說明理由。採用multi-thread的方式，一個thread被阻塞之後，其餘thread仍可動作，優於single-thread，但是，single-thread的context switch次數絕對低於multi-thread

10. (8%) There are four milestones in the developments of Operating System, (time sharing, batch, multi-processor, multi-programming). (a) Please arrange them chronologically, (b) Since which milestone, the context switch time is more sensitive than ever? 四個里程碑，依照時間先後排列，batch --> multi-programming --> time sharing --> multi-processor 哪個系統開始context switch time受到重視？time sharing
11. (15%) There are three queues (Q0, Q1, and Q2) in the system, the priority of Q0 is the highest, and then Q1, and then Q2. The scheduling of Q0 and Q1 are RR, with time quantum 3(Q0) and 6(Q1), whereas Q2 follows the Shortest Remaining Time strategy. When a process is ready to run, it is moved into Q0. Exhausting time-quantum demotes a process to a lower priority queue, i.e., Q0 to Q1 and Q1 to Q2. Arrival time and CPU burst time are shown in the table. Please draw a Gantt chart to demonstrate the selection of queues as well as processes. Calculate the waiting time for each process. 畫出Gantt chart，圖中需標明executing process以及其所屬的queue。算出每一個行程的等待時間。

Process	Arrival	CPU Burst
A	0	19
B	0	2
C	0	13
D	4	12
E	7	8

<參見excel 檔案>

Process	Arrival	End	Execution	Turnaround	Waiting
A	0	54	19	54	35
B	0	5	2	5	3
C	0	44	13	44	31
D	4	40	12	36	24
E	7	37	8	30	22

12. (20%) Explain the following terms. Please demonstrate your knowledge in associate with those terms, instead of translations. Examples are very welcome. Clear description helps credit. (a)background process (b)monolithic kernel (c)dispatch latency (d)exponential average
解釋下列名詞。請展現對於作業系統的理解程度，切勿翻譯了事。盡可能多用例子。字數太少，資訊不足都很可能導致無法正確的評估。請盡量多寫，寫清楚。